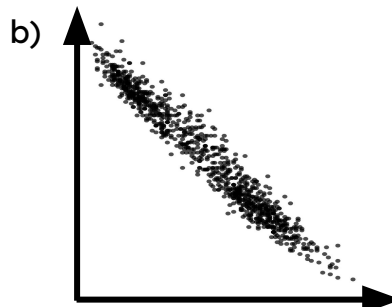
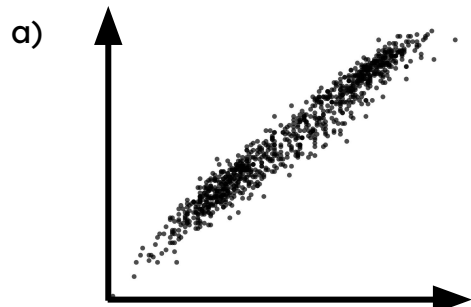




Correlation

Task A: Recognising correlation in bivariate data

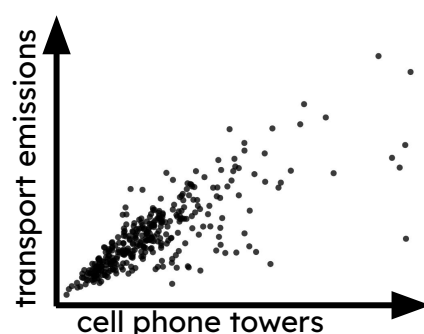
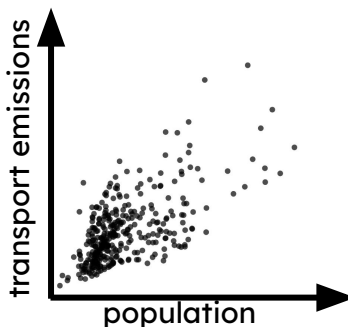
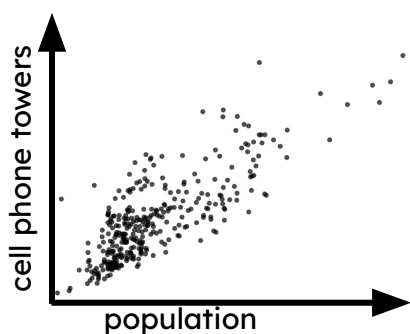
1) Fully describe the **correlation** in each scatter graph.



2) For each set of bivariate data, tick whether you would expect the **correlation** to be positive or negative.

	positive	negative
children's heights and their shoe sizes		
steepness of a hill and walking speed up the hill		

3) The three scatter graphs show data from the Office for National Statistics. Each point represents a region in England and Wales in 2015.



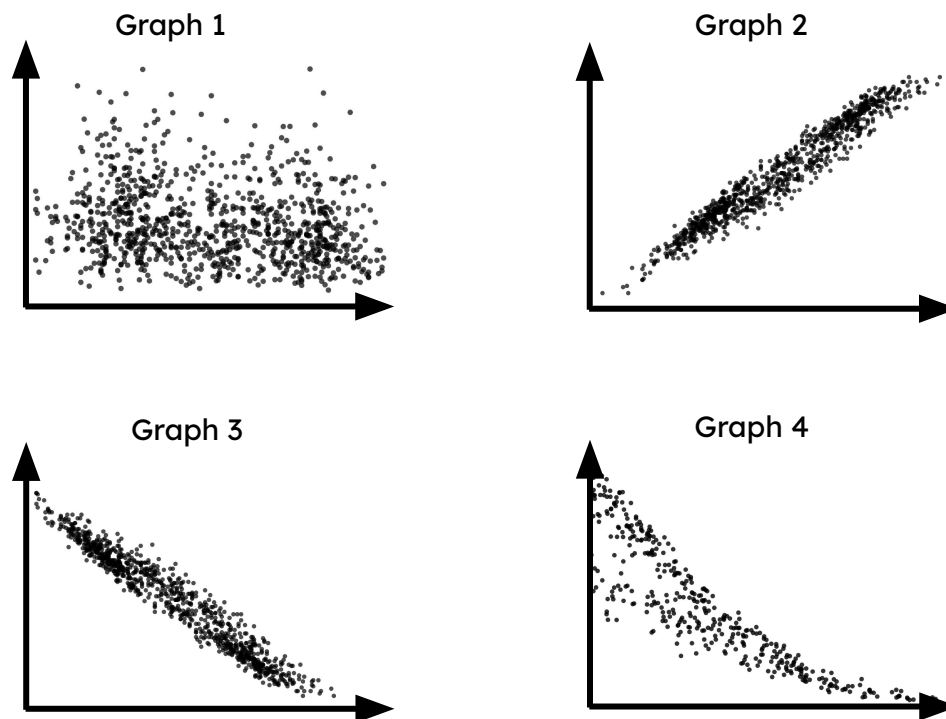
Jacob thinks that the third scatter graph shows that cell phone towers cause more traffic pollution because there is a positive correlation.

Explain why Jacob could be wrong.



Task B: Interpreting bivariate data without correlation

1) Here are four scatter graphs.



- a) Which graph shows a **positive correlation**?
- b) Which graph shows a **negative correlation**?
- c) Which graph shows an association which is not linear?
- d) Which graph shows no association?

